FAIRification and Institutional Linkages in Research: A Publication-Centred Analysis of Dimensions-Recorded Grant and Policy-Document Links

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Abstract

FAIRification is increasingly discussed through the linked concerns of metadata, semantic interoperability, research-data stewardship, Open Science, and library support. Yet the institutional relations surrounding this literature are less often examined in a design that keeps text-based thematic structure separate from database-recorded grant and policy links. This study begins with a curated 163-record library-focused FAIRification source cohort derived from a Scopus study and constructs a 144-publication exact-DOI Dimensions anchor. A publication-only nonnegative matrix factorization of titles and available abstracts is fixed before relational retrieval. The selected three-component representation identifies institutional stewardship, research-data management, and open-science practice; technical metadata, semantic infrastructure, and reproducible workflows; and digital cultural heritage, collections, and preservation. Among the 144 anchor publications, 52 have a Dimensions-recorded grant link and four have a recorded policy-document link. In a publication-year-adjusted model, ten-percentage-point compositional shifts from the technical-metadata reference toward either institutional stewardship/open science or heritage/preservation are associated with lower odds of a recorded grant link. The role-separated institutional landscapes contain 313 stored publication-affiliation organizations and 47 grant-funder organizations; country memberships are non-exclusive stored attributes. Policy-link models are deliberately withheld because the primary observation window contains only three events, below the predeclared adequacy threshold. A publication-year-stratified endpoint-permutation test does not show excess grant–policy co-location on the same publications. The results concern text structure and Dimensions-recorded identifier relations. They do not establish causal funding effects, policy endorsement, uptake, impact, or institutional influence.

1 Introduction

The FAIR principles have become a widely used vocabulary for the stewardship of digital research objects. Their original formulation concerns the extent to which data and metadata can be found,

accessed under stated conditions, interoperated with, and reused by people and machines [13]. FAIR is intentionally principles-level and technology-agnostic. It is not a prescribed implementation standard, a binary certification, or a synonym for unrestricted open access. This breadth supports uptake across disciplines, but it also means that FAIRification is realised through multiple workflows, infrastructures, and institutional arrangements.

The library-focused literature mapped by Papachristopoulos et al. [10] is an especially useful setting for investigating that heterogeneity. Its Scopus analysis identifies metadata, interoperability, data management, and Open Science as central directions while showing that library-related FAIR scholarship occupies a comparatively limited but distinctive position in the broader discussion. Libraries can contribute research-data services, training, repository support, metadata practice, curation, and preservation, but they operate within a wider ecosystem of researchers, infrastructure providers, funders, policy organisations, and disciplinary communities. A bibliometric field map does not by itself show how those actors are connected in recorded institutional data.

This paper therefore uses a publication-centred relational design. It preserves the source cohort rather than defining a new keyword corpus in Dimensions. It then creates a high-confidence Dimensions publication anchor through exact DOI-first matching. Thematic analysis of publication titles and available abstracts is completed before grant and policy-document outcomes are retrieved or examined. After the thematic representation is frozen, the study describes three Dimensions-recorded relation layers: publication–grant (P–G), publication–policy-document (P–D), and policy-issuer–publication. The design makes it possible to distinguish publication-text structure from the database relations attached to publications.

The paper asks four questions:

RQ1. Can the Dimensions-recovered FAIRification anchor be represented by a small, interpretable, and diagnostically supported publication-text structure?

RQ2. Are continuous shares of the resulting themes conditionally associated with a publication's recorded grant-link status after adjusting for publication year?

RQ3. What descriptive scope, timing, role-separated organization/country landscapes, and network properties characterise the recorded grant and policy-document relation layers?

RQ4. Do recorded grant and policy-document relations co-locate on the same publications more often than expected under a publication-year-stratified permutation null?

The answers are deliberately bounded. A Dimensions-recorded grant link is not evidence that a funder selected, caused, or fully financed a publication. A policy-document link is not evidence of policy uptake, endorsement, influence, or impact. Stored record years are descriptive database dates, not causal lags. These limits are central to the analysis rather than qualifications appended after the fact.

2 Literature review

2.1 FAIRification as a context-dependent implementation problem

FAIR supplies high-level guidance for machine-actionable data and metadata, but it does not select a technical stack, repository, vocabulary, or governance model [13]. Its non-prescriptive

character makes it broadly applicable while creating room for divergent implementations. This is why FAIRification should be understood as an implementation process rather than as a single technical act. A generic workflow includes analysis of data and metadata, semantic modelling, transformation to machine-readable and linkable representations, hosting, and assessment [8]. A more recent framework similarly treats FAIRification as iterative and project-specific, with objectives that may justify a context-sensitive "FAIR enough" endpoint rather than maximal compliance on every criterion [12].

Assessment does not remove this contextual dependence. The FAIR Data Maturity Model was developed because varied interpretations can make unharmonised FAIR assessments difficult to compare [1]. Maturity indicators can support transparent improvement, but they do not prove data quality, scientific validity, interpretability, ethical adequacy, successful reuse, or computational reproducibility. The present study consequently examines the scholarly and institutional organisation of FAIRification research; it does not score the FAIRness of individual datasets or infer real-world implementation success from publication counts.

2.2 Libraries, stewardship, and research infrastructures

The source study places libraries in the FAIRification ecosystem because FAIR-oriented practice requires durable work with metadata, documentation, curation, preservation, repositories, and researcher support [10]. Institutional case studies show how libraries can translate such responsibilities into data-management-plan training, consultancy, service catalogues, and repository guidance [6]. A broader systematic review likewise characterises academic-library research-data-management support as a combination of services, education, infrastructure, workflows, and partnerships across the research-data life cycle [7]. These studies support a stewardship perspective, but they do not show that libraries alone determine FAIR adoption or that a particular service causes FAIRness or reuse.

The infrastructure setting is wider than any one institution. FAIR-related interoperability can concern datasets and metadata as well as services and e-infrastructures, with technical, semantic, organisational, and legal dimensions [3]. This provides a substantive rationale for expecting FAIRification research to comprise different orientations: institutional stewardship and Open Science practice; technical metadata and semantic-workflow development; and domain-oriented work around collections and preservation. It does not imply that those orientations are mutually exclusive at the publication level.

2.3 Recorded funding and policy relations as bounded evidence

Publication-linked funding data can be valuable for describing research relations, but their coverage, extraction, and disambiguation limits must be made explicit [5]. A recorded relation is not a complete account of financing, nor does it demonstrate that a grant caused a particular publication. The appropriate empirical question here is therefore associational: which FAIRification text orientations are differentially represented among publications with a Dimensions-recorded grant link?

Similar caution applies to policy documents. Policy-document mentions or citations can identify a detectable relation between a policy-related document and a scholarly publication, but provider coverage and the meaning of a mention are heterogeneous [2, 11]. An observed policy link is not a direct measure of societal impact or policy change. This study accordingly treats P–Das a database-recorded layer and uses policy issuers only as descriptive context. Its primary structural test asks

whether P–Gand P–Dco-locate beyond a year-conditioned null; it does not posit a grant-to-policy pathway.

3 Materials and methods

3.1 Source cohort and Dimensions matching

The starting point was the 162-record library-focused export associated with the Scopus study of Papachristopoulos et al. [10]. The supplied export omitted the foundational paper by Wilkinson et al. [13]. A documented manual addition created a frozen 163-record source cohort. The addition is retained as a source-curation decision rather than silently changing the original export.

The Dimensions enrichment used a deterministic DOI-first procedure. Of 163 source records, 148 contained a source DOI. Dimensions returned candidates for 144 of those DOI-bearing records, and all 144 accepted candidates were exact DOI matches. Four DOI-bearing source records did not return a candidate; 15 source records lacked a DOI. Thus, the analytic publication anchor contains 144 records (88.344% of the frozen cohort), while all 19 unresolved source records remain visible in the matching audit. Dimensions is used as an enrichment source rather than as a replacement definition of the Scopus field. The retrieval and relation fields follow the Dimensions Search Language data model [4].

3.2 Outcome-blind publication-text representation

For each of the 144 matched publications, the title was given modest additional weight by repeating it once and was then combined with the available abstract. Abstracts were available for 140 publications. Generic FAIR retrieval anchors (including *FAIR*, *findable*, *accessible*, *interoperable*, and *reusable*) and generic scholarly terms were removed so that the model would not merely reproduce the retrieval language. The resulting text was represented with TF–IDF features (2,275 observed features).

Non-negative matrix factorization (NMF) was used to identify additive, non-negative text components [9]. Candidate solutions with $K = 2$, $K = 3$, and $K = 4$ components used the same preprocessing and a fixed random seed. Each solution was assessed with eight 80% publication-bootstrap replicates. Bootstrap components were matched to the full-corpus components by cosine similarity, and assignment agreement was calculated after matching. Labels were assigned after inspection of terms and diagnostic material, before relational variables were read.

The selected $K = 3$ representation comprises: (1) institutional stewardship, research-data management, and open-science practice; (2) technical metadata, semantic infrastructure, and reproducible workflows; and (3) digital cultural heritage, collections, and preservation. The $K = 2$ solution remains a coarsened sensitivity. The $K = 4$ solution adds a very small, unstable component and is not used as the primary representation.

3.3 Recorded relation layers and descriptive scope

Let $\mathcal{P}$ denote the 144-publication anchor. A publication–grant edge belongs to P–Gwhen Dimensions records a supporting-grant identifier on a publication in $\mathcal{P}$. A publication–policy-document edge belongs to P–Dwhen a Dimensions policy-document record is linked to a retained publication.

Policy-issuer–publication relations are derived descriptively from issuer attributes on linked policy documents. The analysis contains 163 P–Gedges from 52 linked publications to 152 grants, 223 P–Dedges from four linked publications to 222 policy documents, and 47 unique issuer–publication edges involving 44 policy issuers. These are identifier relations recorded by the data source, not asserted substantive ties.

For publication i, the binary grant outcome Y_i^G equals one if at least one P–Gedge is recorded, and the binary policy outcome Y_i^D equals one if at least one P–Dedge is recorded. Overall and supplementary hard-assignment proportions are reported with Wilson score intervals [14]. The hard assignments are descriptive; the primary association model uses continuous NMF shares.

3.4 Role-separated institution and country landscapes

The institutional and country analysis uses explicit stored Dimensions organization identifiers and preserves the role of each identifier. The publication-affiliation view links each retained publication to its stored affiliation organizations (P–PI). The grant-funder view follows a retained publication through its recorded P–G edge to the stored funder organizations of the linked grant (P–G–GI). The policy-issuer view follows a retained publication through its recorded P–D edge to the issuer organization recorded for the linked policy document (P–D–DI). Organization names are descriptive attributes, not matching keys: affiliations, funders, and issuers are never merged by name or across roles.

Stored country attributes are treated as non-exclusive. A publication can contribute to more than one country when its associated organization records more than one country, and no country normalization, regional grouping, or country-level inference is applied. We report the full anchor and the three frozen dominant-$K = 3$ views. For the landscape figures, the eight most represented stored countries or organizations are selected separately within the all-anchor affiliation and funder views before their percentages are displayed across the three theme groups. Policy-issuer landscapes are not plotted because only four retained publications have an issuer path; their scope remains a descriptive result, not evidence of absent policy documents or issuers.

3.5 Association and timing analyses

The primary grant model is a binomial generalised linear model with HC3 robust standard errors. Topic 2 is the compositional reference. The predictors are ten-percentage-point shares of Topics 1 and 3 and centered publication year:

$$\operatorname{logit}\{\Pr(Y_i^G = 1)\} = \beta_0 + \beta_1 S_{i1}^{(10)} + \beta_3 S_{i3}^{(10)} + \gamma(\text{year}_i - \overline{\text{year}}).$$

The model is conditional and descriptive; it does not identify causal effects. Two predeclared grant sensitivities exclude 11 low-margin NMF assignments and add log-transformed citation counts, respectively.

Policy models were subject to a predeclared adequacy rule of at least ten events per non-intercept parameter. The primary policy window includes publications through 2022, allowing at least five calendar years of observation through the 2026 policy-data end year. The longer-window sensitivity includes publications through 2021. If the event rule fails, the specification is recorded as `not_fit`; no sparse or penalised model is substituted to manufacture an estimate.

For timing context, the study calculates grant-start year minus publication year for P–Gedges and policy-document year minus publication year for P–Dedges when both stored years are available. These are record-year differences, not time-to-impact or causal-lag measures.

3.6 Local cross-relation configuration test

The local configuration analysis fixes the observed P–Glayer and permutes the publication endpoints of P–Dties within publication-year strata. The procedure preserves policy-document endpoints, P–Ddegree bundles within each stratum, and all P–Gties. It uses 1,000 permutations and evaluates two predeclared primary statistics: the number of publications with both relation types and the sum of publication-level P–G-by-P–Ddegree products. An unrestricted endpoint permutation is reported only as a labelled sensitivity. A policy issuer is not a third rewired layer.

4 Results

4.1 Scope, relation layers, and thematic structure

Table 1 reports the cohort and relation-layer scope. The 144 publication anchor is a matched analytic subset, not a claim that the remaining 19 source records are absent from Dimensions. Of the final anchor, 52 publications have a recorded grant link (36.111%; 95% Wilson interval 28.721–44.223%) and four have a recorded policy-document link (2.778%; 1.085–6.924%). Figure 1 makes the resulting relation layers explicit.

Table 1: Source-to-Dimensions scope and recorded relation layers.

Quantity	Count	Interpretation
Frozen curated source cohort	163	162-record source export plus documented addition
Exact DOI-matched Dimensions anchor	144	88.344% of curated cohort
Unresolved source records retained in audit	19	Not classified as absent from Dimensions
Grant-linked publications	52	36.111% of anchor
Policy-linked publications	4	2.778% of anchor
Publication–grant edges	163	52 publications; 152 grants
Publication–policy-document edges	223	4 publications; 222 policy documents
Policy-issuer–publication edges	47	44 issuers; descriptive context
Publication–affiliation organization incidences	392	128 publications; 313 stored organizations
Publication–funder representation pairs	101	52 publications; 47 stored funder organizations

The primary $K = 3$ solution has hard-assignment sizes of 51 (35.4%), 80 (55.6%), and 13 (9.0%) publications for Topics 1–3, respectively. Its mean matched bootstrap component cosine is 0.931 and its mean dominant-assignment agreement is 0.935. Topics 1 and 2 are highly stable across the eight bootstrap comparisons. Topic 3 is semantically coherent but has a weaker minimum matched-component cosine (0.039) and one low correspondence; its size and component-specific uncertainty therefore require caution. Supplementary Table 5 provides the full candidate-solution diagnostics.

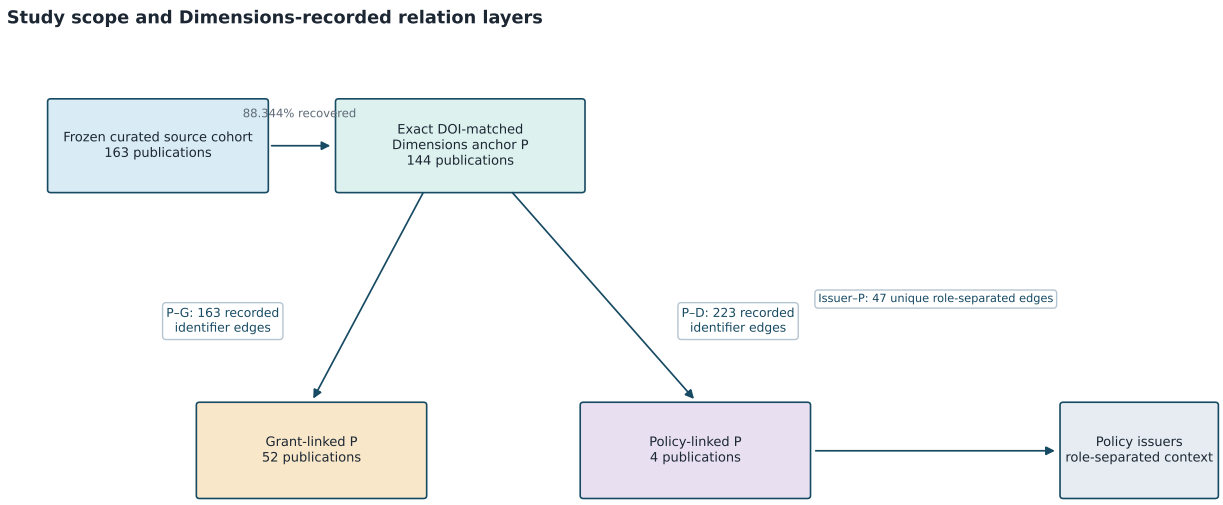


Figure 1: Study scope and Dimensions-recorded relation layers. The frozen source cohort comprises 163 publications, of which 144 exact DOI-matched publications form the final Dimensions anchor. All displayed lines are database-recorded identifier relations and are not measures of causality, funding intent, policy endorsement, uptake, or impact.

Figure 2 shows descriptive hard-assignment linkage rates. The technical metadata/workflow theme has 40 grant-linked publications among 80 (50.0%; 39.297–60.703%), compared with 10/51 (19.608%; 11.015–32.458%) for institutional stewardship/open science and 2/13 (15.385%; 4.326–42.235%) for heritage/preservation. All four policy-linked publications are hard-assigned to the technical theme. These policy rates are descriptive because the event count is too low for inferential theme comparisons.

4.2 Recorded grant links

The primary publication-year-adjusted grant model includes all 144 anchor publications and 52 recorded grant-link events. Both non-reference compositional shifts are negatively associated with a recorded grant link. A ten-percentage-point shift toward institutional stewardship/open science rather than technical metadata/workflows has an odds ratio of 0.772 (95% CI 0.684–0.871, $p < 0.001$). The corresponding shift toward heritage/preservation has an odds ratio of 0.785 (95% CI 0.647–0.952, $p = 0.014$). The year coefficient is also negative (OR 0.849, 95% CI 0.727–0.992, $p = 0.039$).

The direction and statistical compatibility of the two theme-share terms persist in both predeclared sensitivity analyses. Excluding 11 low-margin assignments yields ORs of 0.761 and 0.765; adding log citation counts yields ORs of 0.789 and 0.799. Table 2 and Figure 3 report the primary estimates. They are conditional associations with detected grant links, not estimates of funding selection or funding effects.

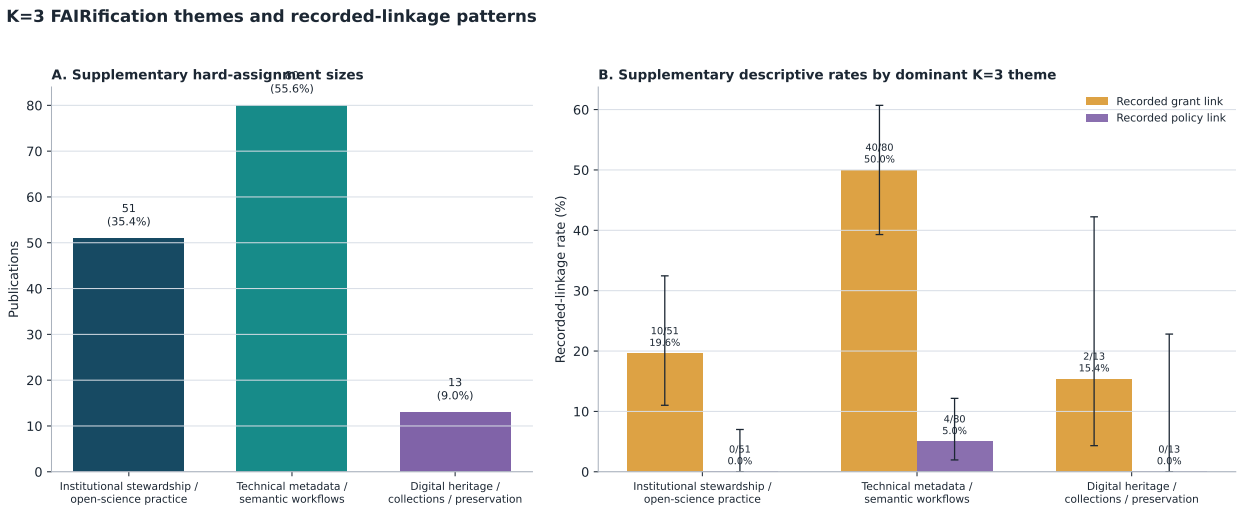


Figure 2: Supplementary $K = 3$ theme linkage rates. Hard-assignment groups and descriptive recorded-linkage rates are shown with Wilson 95% intervals. Primary association analyses use continuous NMF shares. Sparse policy-link counts do not support a multivariable policy model.

Table 2: Primary year-adjusted association model for a Dimensions-recorded grant link ($N = 144$; 52 events). Topic 2 is the compositional reference.

Term	Odds ratio	95% CI	p value
Topic 1 share, per +10 percentage points	0.772	0.684–0.871	< 0.001
Topic 3 share, per +10 percentage points	0.785	0.647–0.952	0.014
Centered publication year	0.849	0.727–0.992	0.039

4.3 Recorded policy links and record-year context

The policy outcome is sparse by every predeclared observation window. The primary through-2022 window contains three policy-linked publications among 72 eligible publications (4.167%). The longer-window sensitivity contains three events among 55 publications; the all-anchor sensitivity contains four events among 144 publications. Each specification has fewer than 30 events required for three non-intercept parameters. The policy model is therefore reported as `not_fit`, rather than being replaced by an opportunistic sparse regression.

Record-year context reinforces the need for descriptive interpretation. For P–G, 162 of 163 edges have both a publication year and a grant-start year; 156 have a negative stored difference, four have the same year, and two have a positive difference. The median grant-start-year-minus-publication-year difference is −3 years. For P–D, all 223 edges have both stored years; 222 policy-document records have a later stored year than the linked publication, one has the same year, and the median policy-document-year-minus-publication-year difference is +6 years. Neither quantity is a causal lag.

4.4 Role-separated network and local configuration evidence

The recorded grant layer is dispersed relative to the policy layer. Among 52 grant-linked publications, the top 10% of publications account for 48.5% of the weighted P–Gedges; the maximum publication

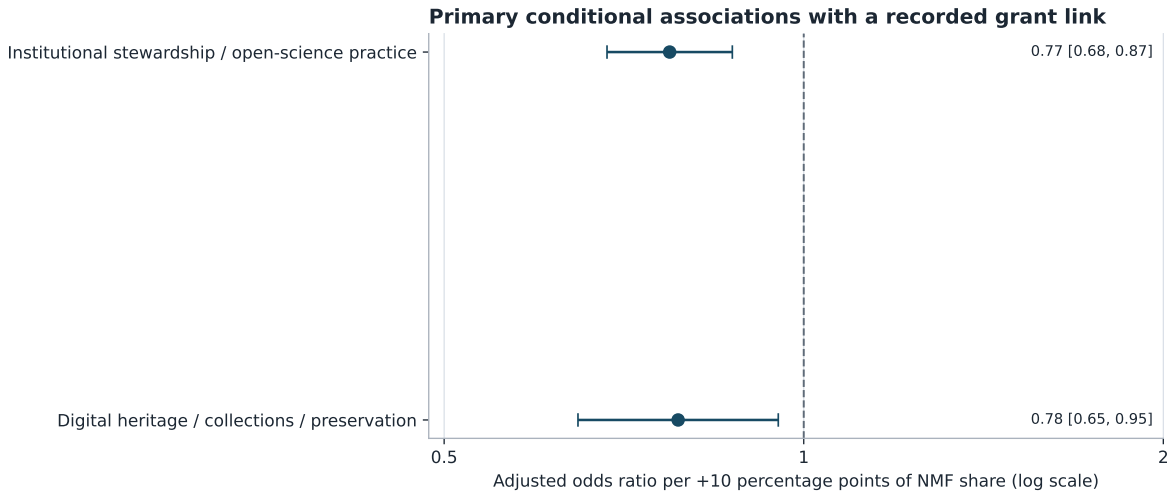


Figure 3: Primary conditional associations with a recorded grant link. Odds ratios are from the predeclared publication-year-adjusted model and are scaled per ten percentage points of NMF share relative to technical metadata/semantic workflows. The estimates are not causal effects.

degree is 19. The policy layer is instead extremely concentrated at the publication side: four policy-linked publications connect to 222 policy documents, and the top policy-linked publication accounts for 97.3% of weighted P–Dincidences. These statistics describe the observed database layer and do not establish prominence, institutional influence, or policy impact.

The four publication configurations are 50 P–G-only (34.7%), two P–D-only (1.4%), two with both links (1.4%), and 90 with neither link (62.5%). Under the primary year-stratified null, the observed number with both links is two, compared with a null mean of 2.229 and a 95% interval of 1–4; its upper-tail permutation p value is 0.822. The intensive degree-product statistic is 3,040, compared with a null mean of 1,619.04 and a 95% interval of 217–3,052; its upper-tail p value is 0.342. Thus, the primary prespecified test does not show enrichment of recorded grant–policy co-location.

The unrestricted endpoint-permutation sensitivity differs: the intensive statistic has an upper-tail $p = 0.018$. This sensitivity does not supersede the primary year-stratified test, because it does not condition on the temporal composition of the P–Dlayer. It is reported to show that the conclusion is null-scheme dependent, not as a post hoc selection of a favourable result.

4.5 Role-separated institution and country landscapes

The affiliation view contains stored organization attributes for 128 of the 144 anchor publications (88.9%): 392 distinct publication–affiliation incidences involve 313 organizations and 46 stored countries (Table 3). The grant-funder view reaches 47 organizations and 17 stored countries through 52 grant-linked publications. Its 101 publication–funder representation pairs are distinct from the 152 linked grants and the 163 raw P–G edges because a publication–funder pair is counted once in this landscape. The policy-issuer view contains 44 organizations and 19 stored countries, but all 47 publication–issuer representation pairs are reached from just four policy-linked publications. It is therefore retained as a role-separated descriptive layer rather than displayed as a third heatmap panel.

The affiliation landscape is geographically and institutionally dispersed. The most frequently

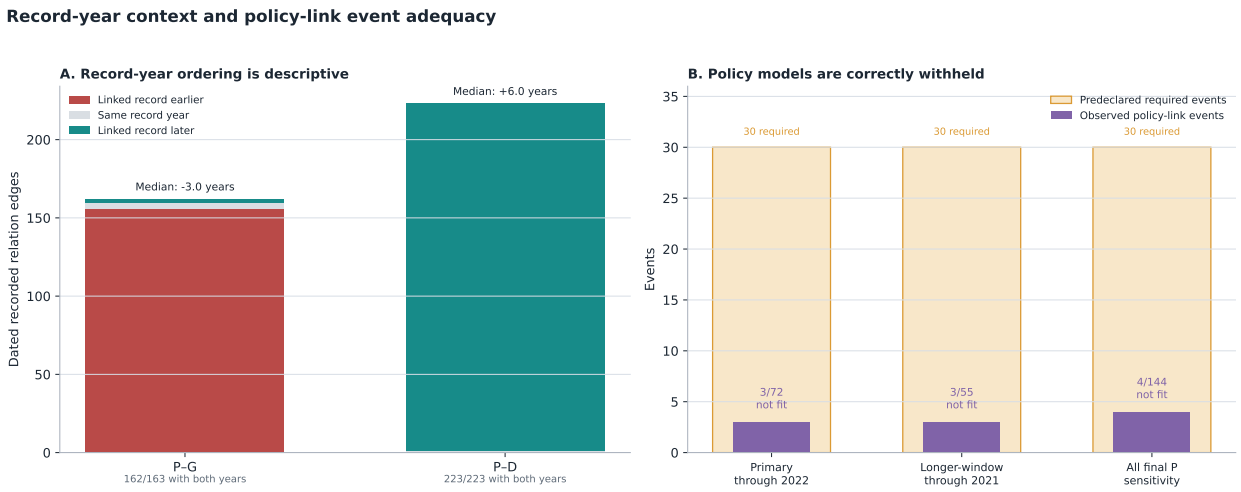


Figure 4: Record-year context and policy-link event adequacy. The timing bars include only relation edges with both stored years; missing years are not imputed. Record-year differences are descriptive database dates. The event panel reports why policy models are withheld under the predeclared 30-event requirement.

Table 3: Role-separated institution and country scope for the final Dimensions anchor. Organization and country counts use explicit stored Dimensions attributes. Publication–target pairs are deduplicated within role; country memberships are non-exclusive.

Role-separated view	Publications with target	Organizations / countries	Representation pairs
P–PI publication affiliations	128	313 / 46	392
P–G–GI grant funders	52	47 / 17	101
P–D–DI policy issuers	4	44 / 19	47

represented stored affiliation countries are the United States (37 anchor publications; 25.7%), Germany (25; 17.4%), the Netherlands (18; 12.5%), and the United Kingdom (17; 11.8%). The German National Library of Science and Technology and Wageningen University & Research each occur as stored affiliations on six anchor publications. The funder view is more concentrated in the technical-metadata/semantic-workflow theme: 40 of its 80 publications reach at least one recorded funder organization, compared with 10 of 51 in Theme 1 and two of 13 in Theme 3. The European Commission (12 anchor publications), the National Institute of General Medical Sciences (8), and the European Research Council (7) are the most represented stored funder organizations. These are database-recorded role attributes and publication representations; they do not rank institutional influence, research capacity, funding allocation, or country effects.

5 Discussion

The study adds an institutional-relational layer to an existing bibliometric starting point without redefining its field through a new Dimensions search. The principal advantage of this approach is provenance: the original library-focused cohort, the documented addition of the FAIR Principles paper, the DOI-first recovery loss, the text-only thematic decision, and the later relational layers remain analytically distinct. The principal cost is that inference applies to the 144 exact DOI-

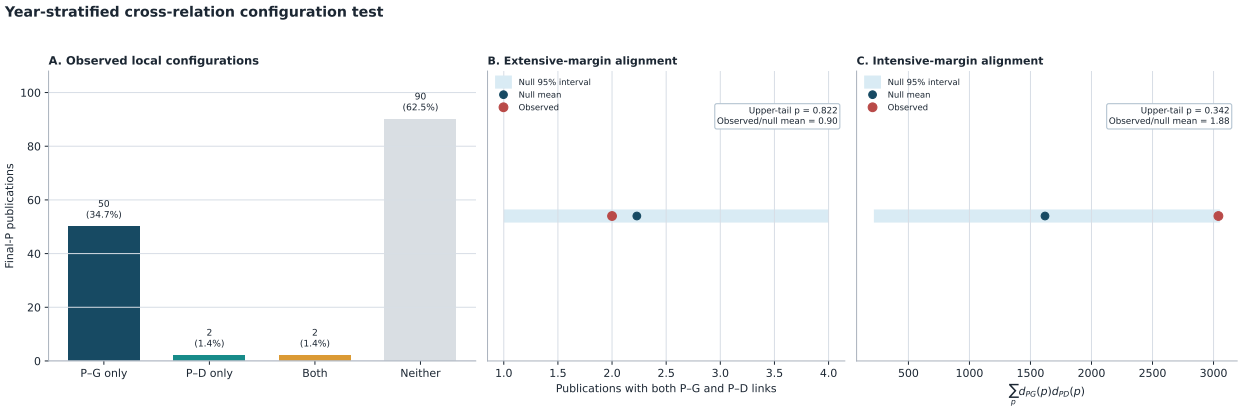


Figure 5: Year-stratified cross-relation configuration test. The primary null reassigns P–Dpublication endpoints within publication-year strata while holding P–Gties fixed. The test concerns structural alignment of recorded layers, not a grant-to-policy causal pathway.

recovered publications rather than automatically to all 163 curated source records.

The thematic results are compatible with the source study's emphasis on metadata, interoperability, research data management, and Open Science [10], but they provide a more operational partition for the matched publication anchor. Technical metadata, semantic infrastructure, and reproducible workflows form the largest component and carry the greatest recorded grant-link prevalence. Institutional stewardship/open-science and heritage/preservation work is less represented among recorded grant-linked publications after adjusting for publication year. This pattern may reflect differences in database coverage, documentation practices, funding acknowledgement, project organisation, publication venues, or the distribution of grants across research activities. It must not be described as evidence that funders favour one FAIRification orientation.

The policy layer is more limited. Its four linked publications and 223 linked policy documents create a visually and numerically concentrated structure, but they do not support a multivariable policy model. The decision not to fit such a model is a substantive safeguard: the available data cannot responsibly distinguish policy-link differences between themes at this event count. More generally, a policy-document relation can identify a candidate research–policy connection while remaining insufficient evidence of influence or impact [2, 11].

The institution and country landscapes add a necessary descriptive distinction to this result. Publication affiliations are widespread across the anchor, whereas recorded funder organizations are reached through only the grant-linked subset and policy issuers through only four publications. The prominence of the European Commission, National Institute of General Medical Sciences, and European Research Council in the funder view, and the representation of the United States, Germany, the Netherlands, and the United Kingdom in the affiliation view, are properties of the stored role-specific records. They cannot establish comparative institutional capacity, country advantage, funder preference, or causal pathways. The theme panels are also hard-assignment descriptions; they complement rather than replace the continuous-share association model, and the small heritage/preservation theme retains its component-specific stability caution.

The primary configuration test also counsels restraint. The observed co-location of grant and policy links is not above the year-conditioned expectation under the prespecified null. The divergent unrestricted sensitivity indicates that temporal constraints matter for the apparent structure.

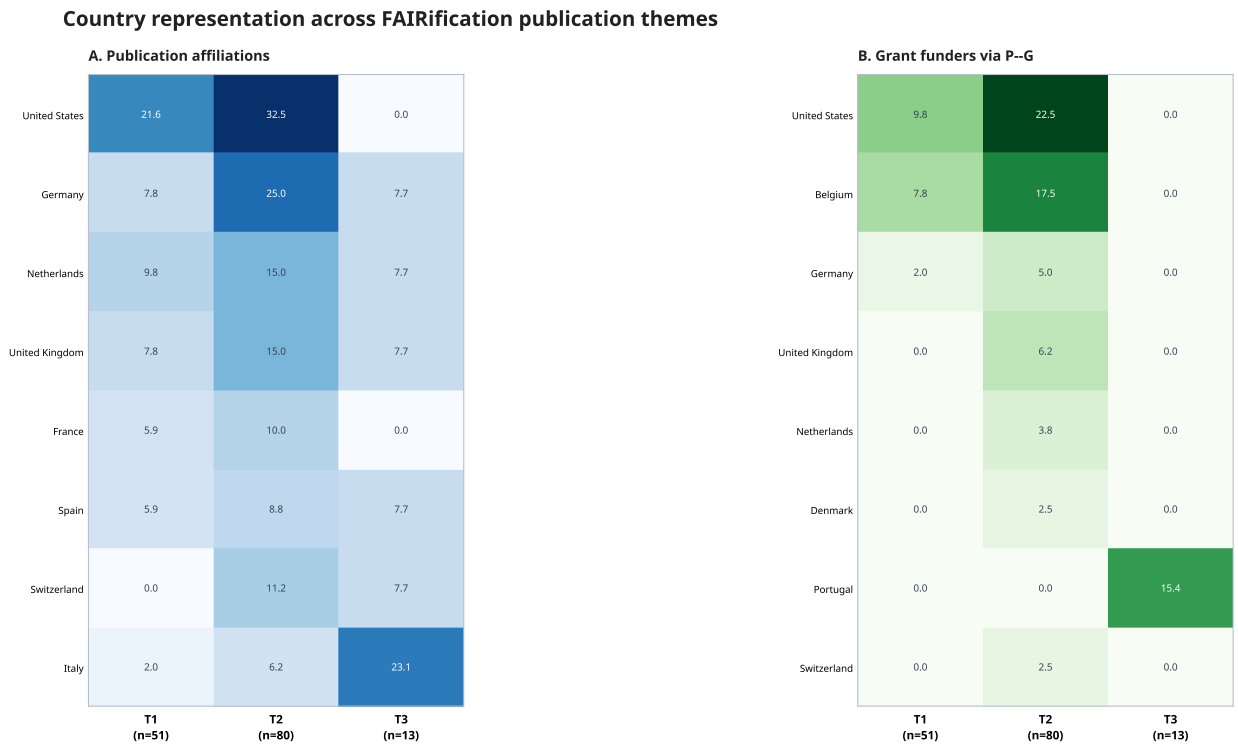


Figure 6: Country representation across FAIRification publication themes. For each of the publication-affiliation and grant-funder views, the eight most represented stored countries are selected from the full anchor and then displayed as the percentage of publications in each dominant $K = 3$ theme with the relevant recorded country attribute. Country memberships are non-exclusive. The policy-issuer panel is omitted because only four retained publications have a recorded issuer path; this display decision is not evidence of missing policy documents or issuers.

Reporting both results is more informative than reducing them to a simple yes/no claim about alignment. Future work could investigate these patterns with broader matched cohorts, validated policy-document typologies, fuller grant metadata, and designs that directly observe funding decisions or policy use.

Several limitations should guide interpretation. First, 19 source records are unresolved under the high-confidence DOI-first procedure. Second, the NMF theme labels are analytical summaries rather than intrinsic categories, and the small heritage/preservation component has a component-specific stability caveat. Third, the study does not claim a K=3 leave-one-out sensitivity for the manual addition; the completed leave-one-out check is limited to the coarsened K=2 structure, where the minimum matched cosine is 0.9998. Fourth, recorded relation fields and organization/country attributes depend on the coverage and entity-resolution behaviour of the data source. Names are not used to merge roles, and stored country memberships are non-exclusive rather than normalized country measures. Finally, the policy layer is too sparse for the intended comparative model.

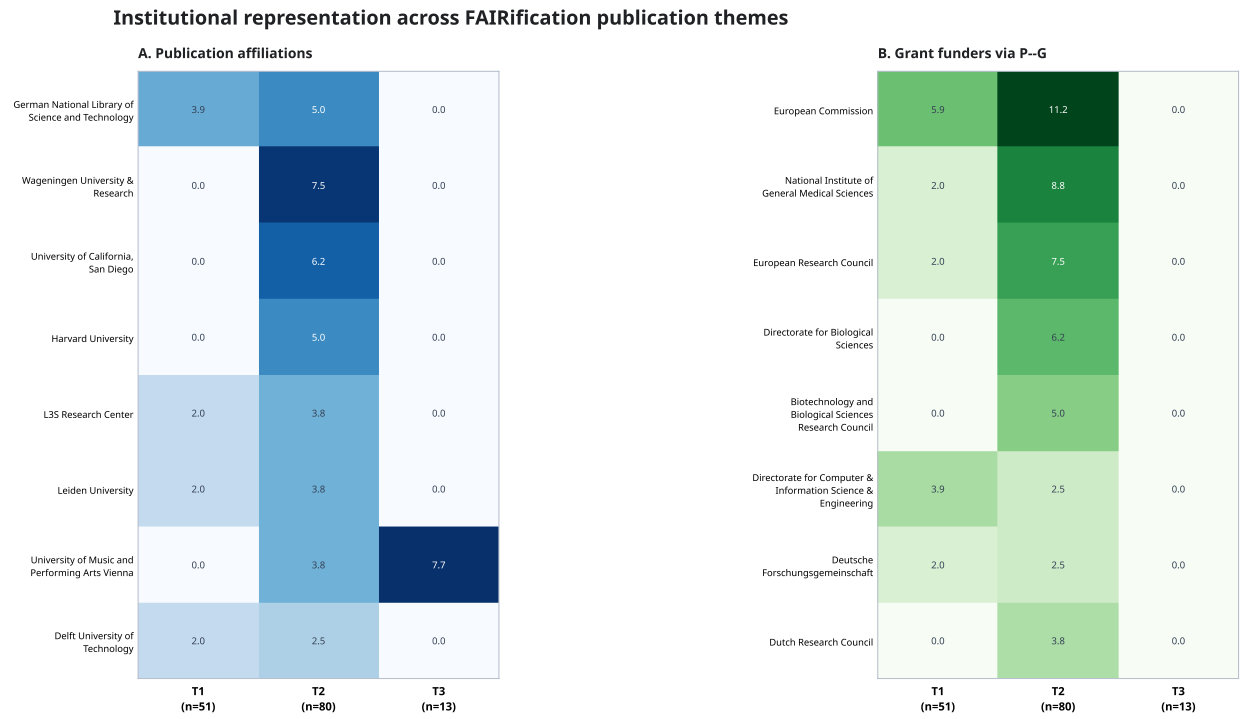


Figure 7: Institutional representation across FAIRification publication themes. For each of the publication-affiliation and grant-funder views, the eight most represented stored organizations are selected from the full anchor and then displayed as the percentage of publications in each dominant $K = 3$ theme with the relevant recorded organization. Organization identities remain role-specific and are not merged by name. The policy-issuer panel is omitted because only four retained publications have a recorded issuer path. These descriptive landscapes do not measure institutional influence, funding intent, policy uptake, or impact.

6 Conclusion

FAIRification research can be organised into at least three publication-text orientations that connect differently to Dimensions-recorded grant links: institutional stewardship/open-science practice, technical metadata/semantic workflows, and digital heritage/preservation. The evidence is strongest for a conditional association between the latter two non-reference orientations and lower odds of a recorded grant link. Role-separated affiliation and funder landscapes further show that stored institutional and country representations differ across these descriptive views, while the policy-issuer layer remains too sparse for comparable visualization or modelling. It is not sufficient to estimate thematic policy-link associations or to claim grant–policy alignment beyond the primary year-conditioned null.

The broader contribution is methodological. Starting from a documented external corpus, preserving matching uncertainty, freezing an outcome-blind text representation, and treating relational fields as recorded database links produces a transparent basis for subsequent institutional analysis. It also prevents the most tempting overinterpretations: FAIRification themes are not funding treatments, policy records are not policy impact, and network concentration, stored organization representation, or country membership is not institutional influence.

Code and materials availability

Reproducible code, the transparent analysis protocol, reviewed aggregate-only results, and manuscript figures are available from the public FAIRification and Institutional Linkages in Research GitHub repository. The repository excludes the source cohort, Dimensions-derived record-level material, publication text, identifiers, relation-edge files, and API credentials. Reproducing the retrieval stages requires independently authorised access to Dimensions and compliance with the applicable data-use conditions.

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Supplementary Material

S1. Source curation, matching, and analytic scope

The source study's library-focused Scopus export contains 162 records. The curated project cohort contains 163 after one documented addition of the foundational FAIR Principles article. Of the 163 records, 148 have a source DOI. The DOI-first Dimensions procedure retrieves candidates for 144 records and accepts all 144 as exact DOI matches. Four DOI-bearing source records return no candidate, and 15 source records have no source DOI. The corresponding 19 records remain in the coverage audit and are excluded from the final Dimensions anchor without being declared absent from Dimensions.

Table 4: Supplementary Table S1. Source-to-Dimensions recovery audit.

Audit quantity	Count
Original supplied library-focused export	162
Documented manual addition	1
Frozen curated source cohort	163
Source records with DOI	148
Exact DOI-first Dimensions matches	144
DOI-bearing records without returned candidate	4
Source records without DOI	15
Unresolved records retained in audit	19
Matched publications with abstracts	140
Matched publications without abstracts	4

S2. NMF candidate diagnostics and thematic caution

The text representation contains 2,275 TF–IDF features. All candidate solutions use the same outcome-blind preprocessing, a fixed random seed, eight 80% publication-bootstrap samples, and cosine matching of bootstrap components to the full solution. Table 5 reports the candidate diagnostics. $K = 3$ provides a strong mean stability profile and interpretable components, but its third component has a low minimum cosine in one bootstrap comparison. The component therefore supports a cautious specialist-theme interpretation, not a claim of universal or equally strong stability across all three themes.

Table 5: Supplementary Table S2. Aggregate NMF candidate diagnostics.

K	Reconstruction error	Mean cosine	Minimum cosine	Mean assignment agreement
2	11.576	0.822	0.349	0.826
3	11.473	0.931	0.039	0.935
4	11.391	0.762	0.050	0.835

The documented manual addition was evaluated only for the broad $K = 2$ sensitivity. Omitting it leaves 143 publications. The full-corpus-to-saved-$K = 2$ dominant-assignment agreement is 1.000,

and matched component cosine ranges from 0.9998 to 0.9999. This result does not supply a $K = 3$ leave-one-out inference.

S3. Supplementary descriptive and association results

Table 6 gives the descriptive hard-assignment rates that underlie Figure 2. The zero policy-link counts in Topics 1 and 3, and the small denominator for Topic 3, are reasons to avoid theme-level policy inference.

Table 6: Supplementary Table S3. Descriptive recorded-linkage outcomes by dominant $K = 3$ theme.

Theme	n	Grant-linked n (%)	Policy-linked n (%)
Institutional stewardship/open science	51	10 (19.6)	0 (0.0)
Technical metadata/semantic workflows	80	40 (50.0)	4 (5.0)
Digital heritage/collections/preservation	13	2 (15.4)	0 (0.0)

Table 7 reports the predeclared grant-model sensitivities. Both topic-share coefficients retain the same negative direction in every fitted specification. The citation-adjusted specification is a sensitivity analysis, not a causal adjustment that resolves confounding.

Table 7: Supplementary Table S4. Grant-link association sensitivity summary.

Specification	N	Events	Topic 1 OR (95% CI)	Topic 3 OR (95% CI)
Primary year-adjusted	144	52	0.772 (0.684–0.871)	0.785 (0.647–0.952)
Low-margin assignments excluded	133	45	0.761 (0.666–0.869)	0.765 (0.603–0.971)
Citation-adjusted sensitivity	144	52	0.789 (0.699–0.891)	0.799 (0.656–0.973)

S4. Role-separated network summaries

The P–Glayer has 52 connected publications and 152 connected grants across 163 unique edges. The P–Dlayer has four connected publications and 222 connected policy documents across 223 unique edges. A single policy-linked publication accounts for 97.3% of weighted P–Dincidences. Policy issuers are descriptive attributes: 44 connected issuers are linked to the four policy-linked publications by 47 unique issuer–publication edges and 223 policy-document weights.

S5. Institution and country landscape scope

Table 8 records the role-separated landscape denominators by dominant $K = 3$ theme. The affiliation role is available across all three themes. The funder role is most prevalent in the technical metadata/semantic-workflow theme, while issuer representation is confined to the four policy-linked technical-theme publications. These descriptive counts retain the role boundary: P–PI affiliations, P–G–GI funders, and P–D–DI issuers are not merged.

The all-anchor affiliation landscape is led by the United States (37 publications; 25.7%), Germany (25; 17.4%), the Netherlands (18; 12.5%), and the United Kingdom (17; 11.8%). The corresponding

Table 8: Supplementary Table S5. Role-separated organization and country scope by dominant $K = 3$ theme. Each cell reports publications with at least one stored target / distinct organizations / distinct stored countries. Country memberships are non-exclusive.

Theme (n)	P–PI affiliations	P–G–GI funders	P–D–DI issuers
Institutional stewardship/open science (51)	45 / 75 / 22	10 / 9 / 3	0 / 0 / 0
Technical metadata/semantic workflows (80)	73 / 243 / 39	40 / 45 / 16	4 / 44 / 19
Digital heritage/collections/preservation (13)	10 / 17 / 12	2 / 1 / 1	0 / 0 / 0

funder landscape is led by the United States (23; 16.0%) and Belgium (18; 12.5%). These percentages use the 144-publication anchor and are not mutually exclusive country shares. Table 9 reports selected named organization representations that are also displayed in Figure 7.

Table 9: Supplementary Table S6. Selected all-anchor organization representations by role. Counts are distinct associated publications; each percentage uses the 144-publication anchor. Organizations are explicit stored Dimensions IDs within a role and are not name-merged across roles.

Role	Stored organization	Publications	Anchor %
Affiliation	German National Library of Science and Technology	6	4.2
Affiliation	Wageningen University & Research	6	4.2
Affiliation	University of California, San Diego	5	3.5
Affiliation	Harvard University	4	2.8
Funder	European Commission	12	8.3
Funder	National Institute of General Medical Sciences	8	5.6
Funder	European Research Council	7	4.9
Funder	Directorate for Biological Sciences	5	3.5

S6. Policy event accounting and configuration-null details

The policy event-adequacy rule requires at least 30 events for the three non-intercept parameters. The primary through-2022 window has 3/72 events, the longer through-2021 window has 3/55, and the all-anchor sensitivity has 4/144. All are reported as `not_fit`.

For the primary year-stratified configuration null, the extensive statistic (N_{both}) has observed value 2, null mean 2.229, null 95% interval 1–4, and upper-tail $p = 0.822$. The intensive degree-product statistic has observed value 3,040, null mean 1,619.040, null 95% interval 217–3,052, and upper-tail $p = 0.342$. The unrestricted sensitivity has an intensive-statistic upper-tail $p = 0.018$; it is retained as a labelled sensitivity because it does not preserve publication-year strata.